

Pulling Equations Out of a Black Box

Neural-Symbolic Approaches to Improving Energy Predictions in Quantum Chemistry

1. Background

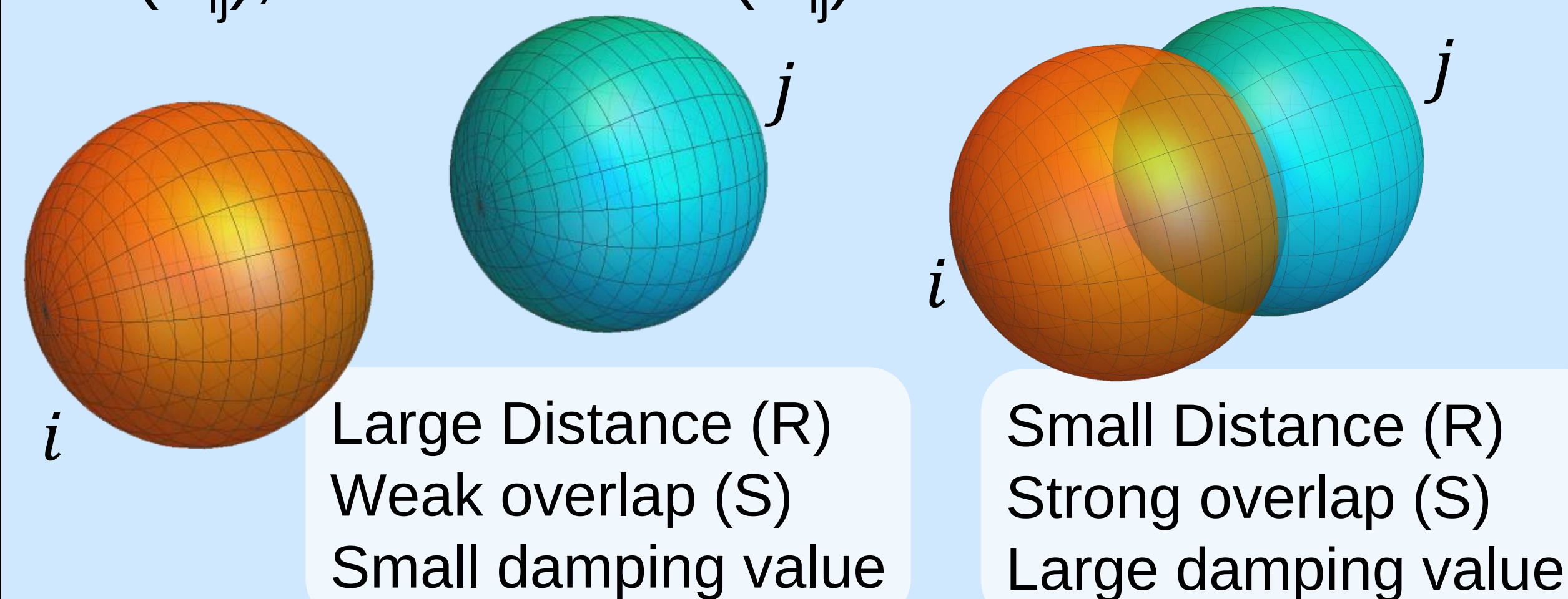
- The Effective Fragment Potential (EFP) method is a computationally inexpensive way of modeling intermolecular interactions.
- Interaction energy in EFP is the sum of: Coulombic effects, dispersion, exchange repulsion, charge transfer, and polarization:

$$E^{int} = E^{Coul} + E^{pol} + E^{exrep} + E^{disp} + E^{ct}$$

- Within EFP, damping functions are used to reduce (dampen) unrealistic energy contributions between fragments at very short intermolecular distances, ensuring correct asymptotic behaviors.

2. Research Questions

- Coulomb energy between fragments i and j must be damped according to their overlap (S_{ij}), and distance (R_{ij}):



$$f^{Damp}(S_{ij}, R_{ij}) = -2 \left(\frac{1}{-2 \ln(S_{ij})} \right)^{\frac{1}{2}} \frac{S_{ij}^2}{R_{ij}}$$

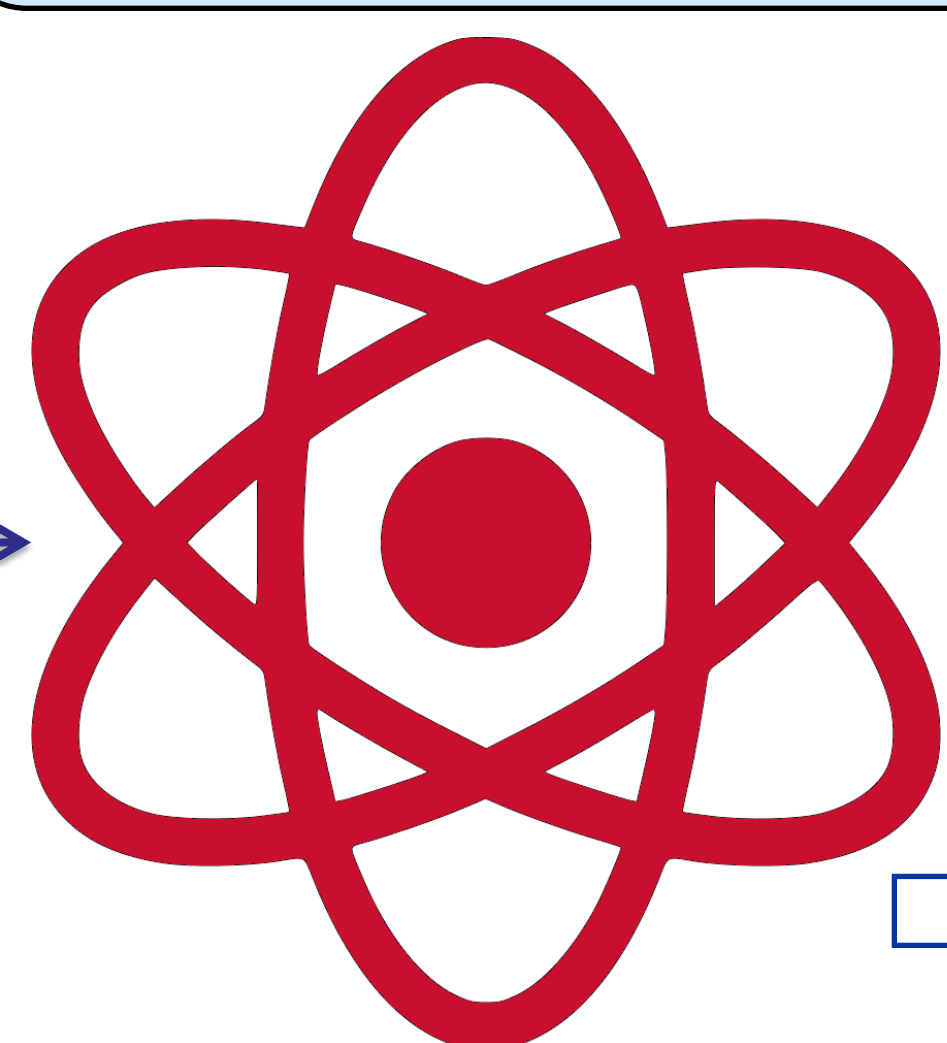
- The total Coulomb correction for a system is:

$$\sum_{j \in B} \sum_{i \in A} f(S_{ij}, R_{ij})$$

- Can an ML data-driven approach derive a Coulomb damping function that is parameter-free (only a function of S_{ij}/R_{ij}), and more accurate than the existing one?

3. A Neural-Symbolic Approach

We attempt to find this new function in 3 phases:



- Train a neural network** to match the per-fragment pair corrections of the theory-derived $f(S_{ij}, R_{ij})$.



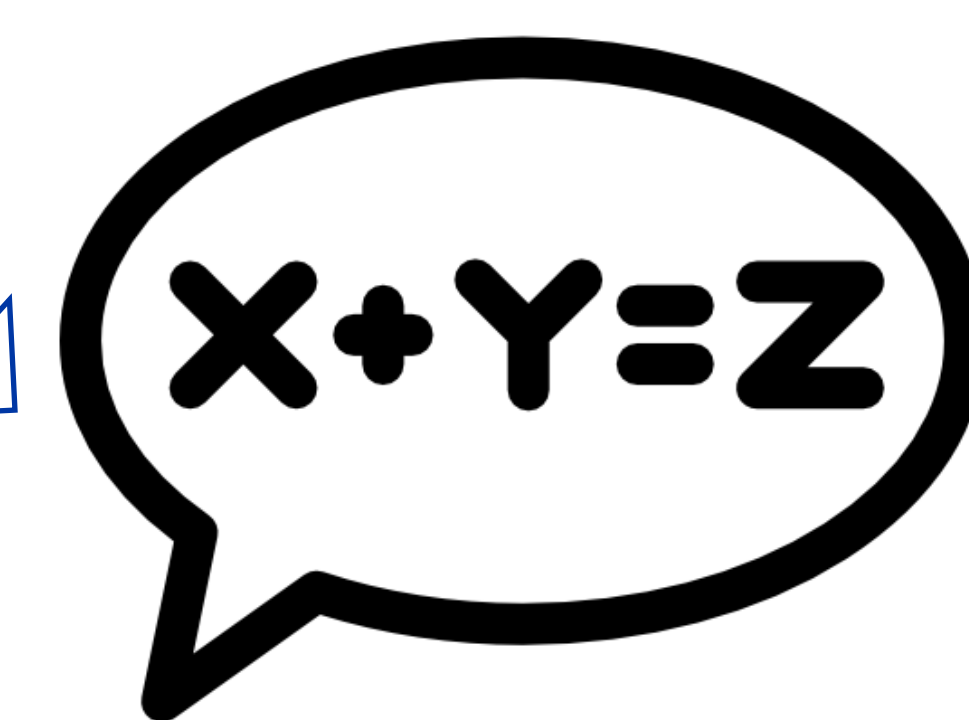
Phase 1: Successful

	$\sum \sum f(S_{ij}, R_{ij})$	NN
Files	466	456
MAE	0.00206	0.00202
RMSE	0.00366	0.00577

- Fine-tune NN** to so that whole-system energy damping corrections approximate results from more accurate calculation methods, like SAPT.



Phase 2: In progress & testing; awaiting more data



- Distill**

interpretable equations from the NN which could potentially replace $f(S_{ij}, R_{ij})$, using symbolic regression.



Phase 3: Awaiting phase 2 completion

4. A Purely Symbolic Approach:

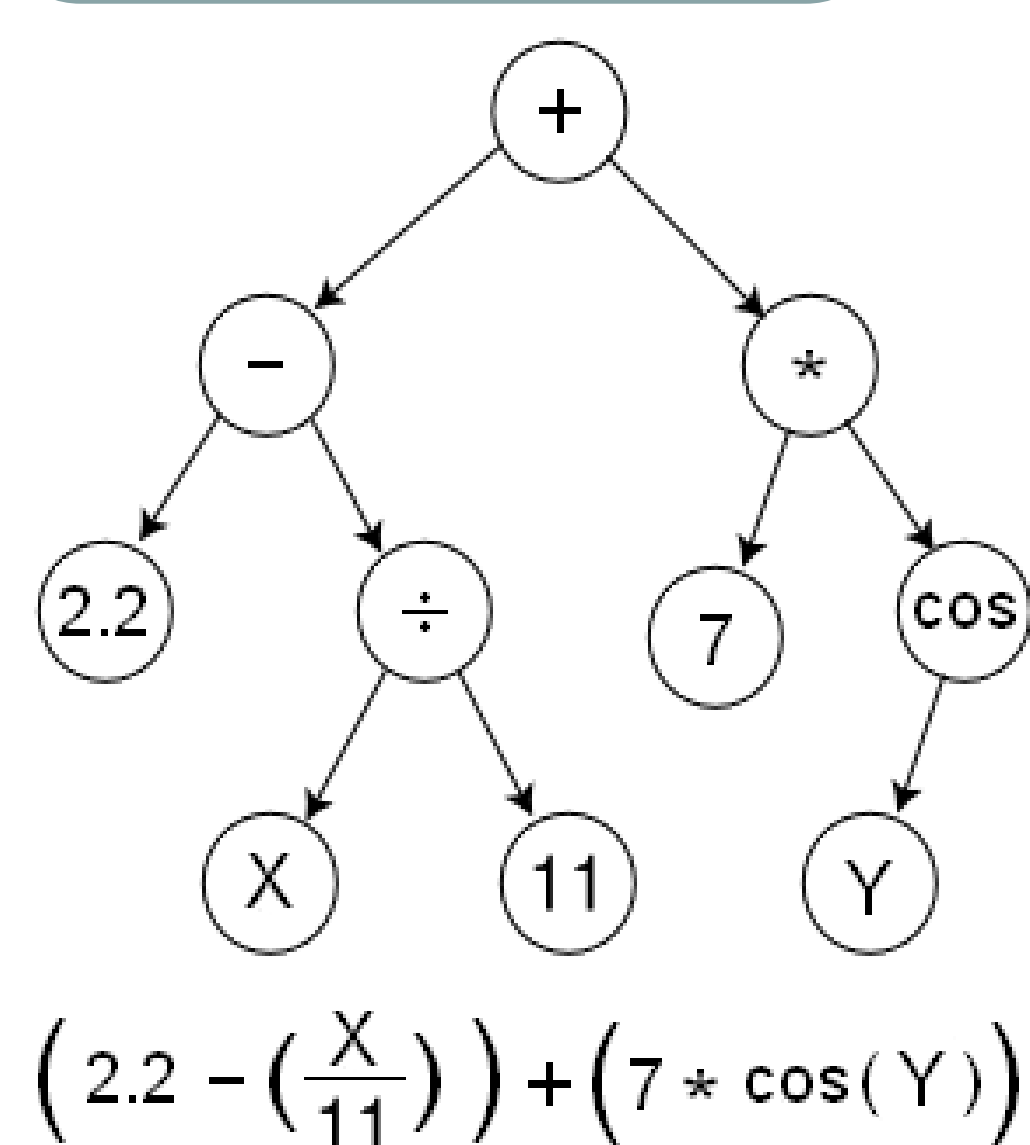
We attempt to find a better function through symbolic regression on our data directly:

Generate thousands of expressions

Evaluate fitness of individuals

Accuracy $\geq$ desired or max # iterations reached?

Return best expressions



Apply genetic algorithms

Add descendants to next generation

5. Symbolic Regression Results

- Best current candidate function (simplicity + accuracy):

$$\text{Systemwise SR Damping Correction} = \sum \sum f(S_{ij}, R_{ij}) - \frac{4 \cdot 10^{-5}}{\sum \sum f(S_{ij}, R_{ij}) + 0.03} E_h$$

	$\sum \sum f(S_{ij}, R_{ij})$	SR Equation
Test Files	119	119
MAE	0.00201	0.00160
RMSE	0.00303	0.00232

- Caveat: Function not differentiable at -0.03; possibly problematic. Preliminary result.

